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Best Practices and Operational Excellence of Well Integrity Surveillance in Oil and Gas Fields

Jaafar Sawad and Hussain Muailu, Saudi Aramco

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Abstract

Maintaining excellent well integrity is a life-cycle obligation extended from drilling to abandonment. Well integrity is a key element in maintaining the safety of fields. Best practices were developed over many years to enhance surveillance practices and deploy new technologies to detect well integrity issues and prolong the life of wells. This paper will discuss integrity surveillance best practices related to casing integrity monitoring, testing wellhead valves and subsurface safety valves (SSSVs).

Casing integrity is a major element of the integrity of the wells. It is being mentioned by close monitoring of the annulus pressures and fluid, running temperature surveys and corrosion logs. The through tubing corrosion logs are the latest technology with the added benefit of providing great insight about the integrity of the wells without the need for removing a completion by a costly workover rig. These logs provide metal loss measures for all casings, utilizing electromagnetic technology. In case a well requires a workover, a number of measures can be taken, such as pressure testing the casings, running logs and repairing leaks by running additional tubulars to maintain the integrity of the well.

The integrity of wellhead valves is another major aspect of well safety. It is associated with preventing uncontrolled releases of formation fluids outside the well. Preventive maintenance and testing for wells are needed through the life cycle of each well. Regular greasing is needed to safely maintain the integrity of valves. Additional frequent testing is needed to detect any faulty valves in a timely manner. In case the valves require replacing, stringent measures are taking place to secure the wells and replace the tree valves.

Last, the paper will go through the SSSV surveillance excellence best practices. Proper testing and troubleshooting of SSSVs is a key element of well integrity. Procedures were developed with a set frequency to test these valves. Troubleshooting is needed in case of questionable testing results. Finally, tubing retrievable SSSVs are converted to wireline valves through rigless operations to maintain the safety of the wells and avoid costly workovers.

Those best well integrity practices are carried out through a well's life cycle to maintain excellent well integrity, protect the environment and optimize the cost of workovers.

Introduction

Well integrity is a critical aspect of oil and gas operations, defined as the ability of a well to contain fluids within the wellbore throughout its lifecycle. This concept is central to ensuring the safety of personnel, protecting the environment, and maintaining the economic viability of hydrocarbon production. Well integrity is not just a technical requirement but also a lifecycle governance issue that demands proactive management from the design phase through to abandonment [3].

The importance of well integrity has grown significantly in recent years due to increasing operational complexities. These include the development of high-pressure high-temperature (HPHT) wells, deeper exploration activities, and the need to maximize production from mature fields. It is important to manage annular casing pressure to ensure the integrity of the wellbore, particularly in onshore and offshore environments [2]. Additionally, detailed equations and calculations for casing, tubing, and drill pipe design are essential for ensuring the mechanical integrity of the well under various operating conditions [1].

This paper focuses on the best practices and operational excellence required to maintain well integrity, with a particular emphasis on well design and installation requirements, casing integrity monitoring, wellhead valve integrity, and SSSV surveillance. A well integrity management system should be implemented to systematically manage risks and ensure compliance with industry standards. [4]. Furthermore, optimized cementing slurry design plays a crucial role in HPHT wells, particularly in formations with salt and water influx, where narrow pressure windows pose significant challenges [5].

Surveillance and monitoring are equally important for maintaining well integrity. This includes regular inspections of wellheads, valves, and seals, as well as monitoring annular casing pressure and the performance of safety systems such as surface safety valves (SSSVs). For instance, API Spec 10a (2019) provides guidelines for well casing and cementing, which are critical for ensuring the integrity of the wellbore. A comprehensive approach to well integrity surveillance, as proposed by Mohammed N Al Khamis et al. (IPTC 17465), helps identify potential risks early and ensures comprehensive monitoring. Similarly, a systematic approach to integrating surface and subsurface well integrity management, as proposed by Karam S. Al Yateem et al. (IPTC 17206), enhances the overall effectiveness of well integrity management.

By adhering to these guidelines, tangible cost savings can be achieved, and retrofit remedial activities can be reduced, such as safety-related workovers, while maintaining the highest safety and operational standards. This paper aims to provide a comprehensive overview of the strategies and best practices for maintaining well integrity, supported by relevant industry standards and case studies, to ensure the safe and efficient operation of oil and gas wells.

Well Design and Installation

The design and installation of oil, water and gas wells are intricate processes that demand meticulous planning and execution to ensure safety, efficiency, and environmental protection. The design and process include:

Material Selection

- **Oil Wells:** Materials must be compatible with hydrocarbons and any potential contaminants. Consideration should be given to the resistance of materials to corrosion, erosion, and fatigue [1].
- **Water Wells:** Materials should resist corrosion from water and any dissolved solids. Consideration should be given to scaling and biological growth [2].
- **Gas Wells:** Materials must be resistant to corrosion from gases such as H₂S or CO₂. Special consideration should be given to high-pressure and high-temperature (HPHT) conditions [3].

Equipment Design

- **Wellhead Equipment:** Designed to withstand the maximum allowable working pressure (MAWP) and operating conditions. All components, including valves, flanges, and seals, must be compatible with the produced fluids [4].
- **Casing and Tubing:** Selected based on the well's depth, pressure, and temperature conditions. Material selection must account for corrosion resistance and compatibility with the reservoir fluids [5].
- **Cementing:** The cementing program must ensure zonal isolation to prevent fluid migration between formations. Cement quality and placement must be verified through logs and pressure tests [6]

Installation

- **Casing and Tubing:** Proper installation techniques must be employed to ensure the structural integrity of the well. This includes proper handling, alignment, and makeup of connections [1].
- **Cementing:** Real-time monitoring of cement placement and quality must be conducted to ensure effective zonal isolation [6].
- **Wellhead:** Installed according to manufacturer specifications and industry standards to ensure safe and reliable operation [4].

Casing Integrity Monitoring

Casing integrity is a fundamental component of well integrity. The casing is a steel pipe that lines the wellbore to prevent collapse and ensure the containment of formation fluids. Over time, casings can degrade due to corrosion, mechanical damage, or thermal stresses, which can compromise well integrity. Therefore, regular monitoring and maintenance of casing integrity are essential.

A systematic processes and activities designed to ensure the structural and functional integrity of the casing strings throughout the well's lifecycle [7]. The casing, which forms the structural backbone of the well, must withstand various mechanical, thermal, and chemical stresses to prevent failure and ensure the containment of fluids within the wellbore. Casing integrity monitoring involves regular surveillance activities, such as annuli surveys, pressure testing, and corrosion assessments, to identify potential issues before they escalate into critical failures [6]. These activities are optimized based on historical data and operational conditions to ensure that the well operates within safe limits while maintaining economic viability.

Adhering minimum surveillance requirements for casing integrity are critical, which include periodic inspections and tests to evaluate the condition of the casing. Advanced technologies, such as logging tools and real-time monitoring systems, play a pivotal role in enhancing the accuracy and efficiency of these surveillance activities. By leveraging these tools, anomalies, such as casing corrosion, cracks, or deformation, can be detected, and corrective actions can be implemented to prevent well failure. Additionally, the integration of CCA (Casing, Cement, and Annulus) Pressure Management Procedures ensures that the casing is not subjected to excessive pressure that could compromise its integrity. The optimization of casing integrity monitoring not only enhances safety but also contributes to significant OPEX savings by reducing the need for costly remedial interventions, such as workovers or well repairs

Annulus Pressure and Fluid Monitoring

One of the most common methods of monitoring casing integrity is by observing annulus pressures and fluid levels. The annulus refers to the space between two concentric casing strings or between the casing

and the wellbore. Changes in annulus pressure or fluid levels can indicate potential issues, such as leaks or fluid migration. For example, an unexpected increase in annulus pressure may indicate a leak in the casing or tubing, while a decrease in pressure could suggest a breach in the annulus.

Regular monitoring of annulus pressures and fluid levels is typically performed using pressure gauges and fluid level sensors. These tools provide real-time data that can be analyzed to identify trends or anomalies. Any deviation from the expected pressure or fluid level should be investigated promptly to prevent further degradation of the casing.

Temperature Surveys

Temperature surveys are another important tool in monitoring casing integrity. Temperature changes in the wellbore can indicate potential issues, such as fluid flow behind the casing or thermal expansion and contraction of the casing. For example, an unexpected temperature anomaly in the annulus could indicate that formation fluids are migrating behind the casing, which could lead to corrosion or pressure buildup.

Temperature surveys are typically conducted using specialized logging tools that measure the temperature profile of the wellbore. These tools can detect subtle changes in temperature that may not be apparent through other monitoring methods. The data collected from temperature surveys can be used to identify areas of concern and guide further investigation or remedial actions.

Corrosion Logs

Corrosion is one of the most common causes of casing failure. Corrosion can occur due to the presence of corrosive formation fluids, such as those containing carbon dioxide (CO₂) or hydrogen sulfide (H₂S) [Over time, corrosion can weaken the casing, leading to metal loss and potentially catastrophic failure [6].

To monitor casing integrity, operators use corrosion logs, which provide detailed information about the condition of the casing. There are several types of corrosion logs, including:

- **Magnetic Flux Leakage (MFL) Logs:** These logs detect areas of metal loss by measuring the magnetic flux leakage from the casing. MFL logs are highly sensitive and can detect even small amounts of metal loss [4].
- **Electromagnetic (EMI) Logs:** These logs use electromagnetic technology to measure the thickness of the casing. EMI logs are particularly useful for detecting corrosion in areas that are difficult to access [6].
- **Acoustic Logs:** These logs use ultrasonic technology to measure the thickness of the casing and detect areas of metal loss. Acoustic logs are highly accurate and can provide detailed information about the condition of the casing [4].

These logs are run through the production tubing, eliminating the need to remove the completion string or perform a costly workover. Through-tubing corrosion logs provide high-resolution measurements of metal loss across all casing strings, allowing operators to identify potential issues before casing integrity is being compromised, resulting in necessity of workover operations to restore the integrity of the well.

Wellhead Valve Integrity

Wellhead valves are critical components of the well system, responsible for controlling the flow of formation fluids and preventing uncontrolled releases. The integrity of wellhead valves is essential to function reliably and maintain a secure seal under various operational conditions. Failure in these valves can lead to safety hazards, environmental damage, and operational downtime. Ensuring the integrity of these valves is, therefore, regular maintenance and testing are imperative to maintain safety of personnel, protecting the environment, and ensuring the economic viability of oil and gas operations [6].

Preventive Maintenance

Preventive maintenance is the cornerstone of wellhead valve integrity. Regular maintenance activities, such as greasing, exercising, and testing, are essential to ensuring that the valves function correctly. Greasing involves lubricating the valve's moving parts to prevent corrosion and wear. Exercising the valve involves opening and closing it to ensure that it operates smoothly and freely. Testing the valve involves verifying its ability to hold pressure and prevent leaks.

Preventive maintenance achieved through the analysis of historical data and operational conditions. By understanding the factors that contribute to valve degradation, such as corrosion, wear and tear, and thermal cycling, operators can implement targeted maintenance strategies to extend the service life of the valves.

Pressure Testing

Pressure testing is a critical component of wellhead valve integrity. Pressure tests are performed to verify the valve's ability to withstand the maximum operating pressure and prevent leaks [8]. There are several types of pressure tests, including:

- **Zero Energy Test:** This test involves closing the valve and applying pressure to the upstream side to verify that the valve holds pressure [4].
- **Full Bore Test:** This test involves opening the valve fully and applying pressure to the upstream side to verify that the valve seats and seals are intact [4].
- **Leakage Test:** This test involves applying a small pressure to the valve and checking for any signs of leakage [4].

Pressure testing should be performed during the initial installation of the valve, as well as at regular intervals throughout the well's lifecycle. Any failure during pressure testing should be investigated and remedied promptly to ensure the integrity of the valve [1].

Defective Valve Replacement

In the event that a wellhead valve is identified as faulty or damaged, replacement becomes imperative to maintain the integrity and safety of the well. Valve replacement is an intricate and critical operation that necessitates meticulous planning and execution. The process involves primary isolation method which are killing the well and plug installation to ensure a safe and successful outcome. Below is the step-by-step procedure for valve replacement:

Killing the Well

Killing the well involves making it safe by preventing any flow of fluids. This is typically achieved by circulating heavy brine fluid into the well to balance the pressure and prevent any influx of formation fluids [1].

Plug Installation

Once the well is confirmed killed successfully a downhole plug is set with wireline unit to have additional barrier. After confirmation of successful plug installation proceed with valve replacement.

Removal and Installation

The faulty valve is carefully removed and replaced with a new one. The replacement valve must be compatible with the existing wellhead equipment and operating conditions to ensure proper function and longevity.

Pressure Testing and Plug Removal

Post-installation, the new valve undergoes Zero Energy test, Full Bore test and Leakage test to verify its functionality and pressure-holding capability. This ensures the valve operates as intended and maintains the integrity of the well. Once all tests successful, wireline unit proceed to remove installed plugs.

Flow back operation

In order to return the well to its normal condition, the pumped kill fluid need to be removed. This can be through normal flow back of the well with build up pressure from reservoir or through nitrogen kick off operation [6],

Safety Measures

Throughout the valve replacement process, stringent safety protocols must be adhered. This includes the use of appropriate personal protective equipment (PPE), adherence to established safety procedures, and ensuring all equipment is in optimal working condition. These measures are essential to safeguard personnel and the environment from potential hazards.

Subsurface Safety Valve (SSSV) Surveillance Excellence

Subsurface safety valves (SSSVs) are critical components of the well's safety system. These valves are installed in the tubing string and are designed to prevent uncontrolled releases of formation fluids in the event of a well control emergency. SSSVs are typically actuated by pressure or fluid flow, and they must function correctly to ensure the safety of the well and the surrounding environment [6].

Testing and Troubleshooting

Regular testing and troubleshooting of SSSVs are essential to ensuring their proper functioning. Testing should be performed at regular intervals,

The testing process typically involves the following steps:

Functional Test. This test verifies that the valve opens and closes correctly in response to the actuation signal.

Pressure Test. This test verifies that the valve can withstand the maximum operating pressure and prevent leaks.

Flow Test. This test verifies that the valve can handle the maximum flow rate without failing.

Any issues or anomalies detected during testing should be investigated and remedied promptly. Troubleshooting may involve analyzing the test data, inspecting the valve, and performing additional tests to identify the root cause of the problem.

Tubing Retrievable SSSVs

Tubing retrievable SSSVs are designed to be part of the completion and can only be retrieved with the tubing. These valves are typically installed in wells where the tubing is frequently pulled for maintenance or replacement. Tubing retrievable SSSVs offer several advantages over conventional SSSVs, including reduced costs, shorter downtime, and improved safety.

In cases where tubing retrievable SSSVs are no longer reliable or require replacement, operators can convert them to wireline retrievable valves through rigless operations. This process involves installing a wireline retrievable valve in place of the tubing retrievable valve, using a wireline unit. The conversion process eliminates the need for a workover rig, reducing downtime and costs.

CCA Pressure Management Procedures

CCA, likely referring to "Casing and Cementing Annulus," involves managing the pressure within the casing and cementing annulus to prevent issues such as blowouts or fluid migration between geological formations. Monitoring and controlling this pressure, ensures the mechanical integrity of the well and preventing sustained casing pressure. These procedures are essential for maintaining the structural integrity of the well and ensuring the safety of operations [5].

Landing Base Inspections

Landing bases are integral components of the wellhead equipment, serving as the points where tubing or downhole tools are secured and sealed. Regular inspections of these areas are crucial to identify any signs of wear, corrosion, or damage that could compromise the seal. The manual outlines guidelines for the frequency of inspections, what to look for, and the necessary actions to take if any issues are detected. This ensures that all connections remain secure and reliable.

Wellhead Seals and Weld Testing Requirements

Wellhead Seals

Wellhead seals are critical in preventing leaks between wellhead components and the casing. Weld testing ensures the integrity of welds in the wellhead and associated equipment, checking for defects that could lead to failures.

Types of Seals. Various types of seals are used, including O-rings, gaskets, and metal-to-metal seals, each selected based on the specific application and operating conditions [4].

Pressure Testing. Seals must undergo pressure testing to ensure they can withstand the maximum operating pressure without leaking. The test pressure is typically at least 80% of the design pressure.

Leakage Testing. Conducted to verify that the seals are intact and functioning correctly. Any detected leaks must be addressed promptly.

Frequency. Pressure and leakage tests should be performed annually or as specified by the well's operating conditions.

Weld Testing

Weld testing ensures the integrity of welds in the wellhead and associated equipment, checking for defects that could lead to failures.

Types of Tests. Visual Inspection: Conducted to identify surface defects such as cracks, porosity, or lack of fusion [8].

Non-Destructive Testing (NDT). Includes methods like ultrasonic testing, radiography, and magnetic particle testing to detect internal defects [4].

Destructive Testing. Involves testing weld samples to their limits to assess strength and durability.

Frequency: Welds should be tested during the initial installation and at regular intervals thereafter, as specified by the well's operating conditions [5].

Regulatory Framework and Industry Standards

Well integrity surveillance is subject to various regulatory requirements and industry standards. These regulations and standards are designed to ensure that wells are operated safely and in an environmentally responsible manner. Some of the key regulatory frameworks and industry standards include:

API Standards

The American Petroleum Institute (API) provides standards for well integrity surveillance, including guidelines for casing design, cementing, and testing.

ISO Standards. The International Organization for Standardization (ISO) provides standards for well integrity management, including guidelines for risk assessment, monitoring, and reporting.

Local Regulations. Operators must comply with local regulations and guidelines, which may vary depending on the authority.

Operators should ensure that all well integrity surveillance activities are conducted in compliance with relevant regulations and industry standards. This includes maintaining accurate records of monitoring and testing activities, as well as reporting any incidents or anomalies to the relevant authorities.

Conclusion

Well integrity surveillance is a critical component of oil and gas operations, requiring careful planning, execution, and monitoring. By adhering to best practices and maintaining operational excellence, operators can ensure the safety of their wells, protect the environment, and optimize well performance. Casing integrity monitoring, wellhead valve integrity, and SSSV surveillance are essential components of well integrity, and their proper implementation is critical to preventing failures and reducing the cost of workovers.

The use of advanced technologies, such as real-time monitoring systems, machine learning, and robotic inspection tools, can enhance well integrity surveillance by enabling early detection of potential issues and reducing the need for costly interventions. Operators should also ensure compliance with relevant regulations and industry standards, and learn from past incidents to improve their practices.

In conclusion, well integrity surveillance is not just a regulatory requirement but a moral and economic imperative. By prioritizing well integrity, operators can ensure the long-term sustainability of their operations and contribute to a safer and more environmentally responsible oil and gas industry.

Recommendations

Based on the discussion above, the following recommendations are made to enhance well integrity surveillance:

- **Adopt Proactive Monitoring:** Operators should adopt a proactive approach to well integrity surveillance by regularly monitoring casing integrity, wellhead valves, and SSSVs.
- **Implement Robust Maintenance Schedules:** Operators should implement robust maintenance schedules for wellhead valves and SSSVs, including regular greasing, exercising, and testing.
- **Ensure Compliance with Regulations:** Operators should ensure that all well integrity surveillance activities are conducted in compliance with relevant regulations and industry standards.
- **Learn from Past Incidents:** Operators should learn from past incidents and use this knowledge to improve their practices and reduce the risk of future failures.
- **Invest in Workforce Training:** Operators should invest in the training and development of their workforce to ensure that they have the skills and knowledge needed to perform well integrity surveillance activities effectively.
- **By implementing these recommendations,** operators can optimize well integrity, reduce the cost of workovers, and ensure the safety of their operations.

References

1. API Standard 5C3. (2019). Technical Report on Equations and Calculations for Casing, Tubing, and Drill Pipe. American Petroleum Institution
2. API RP 90. (2016). Annular Casing Pressure Management for Onshore and offshore Wells. American Petroleum Institution (API)
3. ISO 16530-1. (2017). Well integrity - Part 1: Life cycle governance. International Organization for Standardization (ISO)
4. ISO 16530-2. (2019). Well integrity - Part 2: Well integrity management. International Organization for Standardization (ISO)
5. Muhammad Imran, Mustafa Sarbast, Serwer Yousif, Kamil Shahzad, and Omar El Zanaty Optimized Cementing Slurry Design for HPHT Wells with Salt Formation sand Water Influx Under a Narrow Pressure Window: Case Studies and Best Practices SPE-225000-MS, Society of Petroleum Engineers (SPE).
6. Mohammed N Al Khamis, Faisal Al Khalewi, Mohammed Al Hanabi, Karam Al Yateem, Ayman Al Qatari, Husain Al Muailu. A Comprehensive Approach of Well Integrity Surveillance IPTC 17465. International Petroleum Technology Conference
7. API Spec-10a. (2019). Well Casing and Cementing, Retrieved from <https://standards.globalspec.com/std/14208303/api-spec-10a>. on April 3 2025
8. Karam S. Al Yateem, Husain A. Al Muailu, Samih M. Al Syed and Khalid I. Al Omairen. (2013). Systematic Approach to Integrate Surface and Subsurface Well Integrity Management System IPTC 17206. International Petroleum Technology Conference